

Physical Energy Functionals for Kempe Chain Reconfiguration: Spin Glasses, Magic Gems, and the Four Colour Theorem

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Abstract

We introduce a suite of physical energy functionals for the Kempe chain reconfiguration graph of planar triangulations, drawing on anti-ferromagnetic Potts models, polyhedral covariance (Magic Gems), electrostatic charge flow, and protein folding energy landscapes. By embedding the four colors at the vertices of a regular tetrahedron in $\mathbb{R}^3$ and assigning the fifth color to the centroid, each Kempe (a, b) -swap becomes a 180° rotation in $\text{SO}(3)$, linking the constructive Four Colour Theorem to topological quantum field theory. We define eight energy functionals — including Magic Gem covariance, local entropy, surface tension rigidity, and defect interaction — and evaluate them on the first known counterexamples to BFS-optimal avoidance (Conjecture 5.5) discovered at $n = 9$ vertices. Three principal findings emerge: (1) local entropy cleanly discriminates safe from unsafe reconfiguration paths ($\Delta = +0.271$); (2) Magic Gem energy trajectories expose kinetic trap structure analogous to protein misfolding, with greedy descent leading to merge-prone basins; and (3) surface tension rigidity — the variance of normalized boundary size across Kempe chains — provides a combinatorial signature of merge-prone configurations. We formalize these observations as the Surface Tension Rigidity Conjecture, which we note was subsequently falsified at $n \geq 10$, though the local entropy and kinetic trap signatures persist. We discuss connections to the Penrose evaluation, sheaf cohomology, chromatic polynomials, and the discharging method.

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Code availability: All computational code (Python, NetworkX) is available at <https://github.com/kylemathewson/GraphColour>. The complete enumeration pipeline for planar triangulations through $n = 10$ is in `compute/kempe/`.

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1 Introduction

The Four Colour Theorem (4CT) — every planar graph admits a proper 4-coloring — was first proved by Appel and Haken [1] using an unavoidable set of 1,476 reducible configurations, each verified by computer. A cleaner proof by Robertson, Sanders, Seymour, and Thomas [8] reduced the set to 633 configurations, but the essential reliance on exhaustive computation remains. In the half-century since, no “human-readable” proof has emerged.

A fundamentally different strategy proceeds *constructively*: begin with the easy Five Colour Theorem, then systematically eliminate the fifth color via Kempe chain swaps. This approach has a long and distinguished pedigree — Kempe’s original 1879 “proof” was exactly this strategy, and Heawood’s 1890 counterexample showed only that a particular Kempe swap could merge two chains, not that the entire strategy fails. Mathewson [6] revived and formalized this line of attack. An inductive proof architecture based on vertex removal and lifting achieves a nearly complete proof: six lemmas are proved, covering all cases except when the removed vertex v has degree 4 or 5 and all five colors appear among its neighbors. In this critical case, one must show that the 5-coloring of $G - v$ can be reconfigured — via a sequence of Kempe swaps — into a coloring where two neighbors of v share a color, freeing a color for v .

1.1 The Avoidance Problem

The central obstacle is *avoidance*: during reconfiguration, one must avoid Kempe swaps that merge the two chains incident to v , as such merges destroy the degree of freedom needed to recolor v . Conjecture 5.5 of the constructive architecture asserted that BFS-optimal paths in the reconfiguration graph $\mathcal{R}(G - v, 5)$ always avoid such unsafe swaps. Exhaustive enumeration at $n = 9$ disproves this conjecture: for the triangulations $T_{9,25}$ and $T_{9,35}$, *all* optimal-length paths from certain 5-colorings to the nearest 4-coloring pass through at least one unsafe swap [7].

This disproof, however, is not a disproof of the constructive strategy itself. In every counterexample, *safe* non-optimal paths exist — they simply require one or two extra steps. This led to the revised Conjecture 5.5’ (Safe Path Existence): for every 5-coloring of a planar triangulation and every vertex v , there exists a reconfiguration path to a 4-coloring that avoids all swaps merging chains incident to v .

1.2 Why Physics?

The existence of safe paths raises the question: *what distinguishes safe paths from unsafe ones?* If we view the reconfiguration graph as a discrete energy landscape — each coloring assigned an energy by some functional — then safe and unsafe paths correspond to different trajectories through this landscape. We propose that ideas from statistical physics, specifically anti-ferromagnetic spin models and disordered systems, provide the right language.

The connection is not merely metaphorical. A proper k -coloring of a graph G is precisely a ground state of the anti-ferromagnetic Potts model on G at zero temperature. The chromatic polynomial $P(G, k)$ is the zero-temperature partition function. Kempe chain reconfiguration is the natural dynamics of this system: a Kempe (a, b) -swap flips a connected cluster of spins in the (a, b) -subgraph, which is exactly the Swendsen-Wang cluster update restricted to two colors.

We strengthen this connection by embedding the four colors at the vertices of a regular tetrahedron in $\mathbb{R}^3$, with the fifth color at the centroid. In this embedding, each Kempe swap becomes a 180° rotation in $\text{SO}(3)$, the symmetry group of the tetrahedron maps isomorphically to S_4 , and the entire reconfiguration process becomes a walk through a space of $\text{SO}(3)$ -valued gauge fields on the graph. This connects the constructive 4CT directly to topological quantum field theory (TQFT), and specifically to Kauffman’s reformulation of the 4CT via the Penrose evaluation [4].

1.3 Contributions

In this paper we:

- (i) Define eight energy functionals for the reconfiguration graph, drawn from anti-ferromagnetic Potts models, polyhedral covariance (Magic Gems [7]), electrostatic charge flow, protein folding landscapes, and surface tension (Section 3).
- (ii) Evaluate these functionals on the first known counterexamples to Conjecture 5.5 at $n = 9$ (Section 4).
- (iii) Identify three discriminating signatures: local entropy, Magic Gem energy trajectories, and surface tension rigidity.
- (iv) Formalize the *Surface Tension Rigidity Conjecture*: chains with zero-variance normalized surface tension are merge-prone.
- (v) Discuss connections to the Penrose evaluation, sheaf cohomology, chromatic polynomials, and the discharging method (Section 5).

1.4 Outline

Section 2 reviews Kempe chains, the constructive 4CT architecture, the reconfiguration graph, and the relevant physical models. Section 3 defines all eight energy functionals with the tetrahedral color embedding. Section 4 presents computational results on the $n = 9$ counterexamples. Section 5 develops connections to TQFT, sheaf cohomology, chromatic polynomials, and discharging. Section 6 summarizes and poses open questions.

2 Background

2.1 Kempe Chains and the Constructive 4CT Architecture

Definition 2.1 (Proper coloring). A *proper k -coloring* of a graph $G = (V, E)$ is a function $c: V \rightarrow \{1, 2, \dots, k\}$ such that $c(u) \neq c(v)$ for every edge $(u, v) \in E$.

Definition 2.2 (Kempe chain). Given a proper k -coloring c and two colors $a \neq b$, the *(a, b) -subgraph* is the subgraph of G induced by vertices colored a or b . A *Kempe (a, b) -chain* is a connected component of this subgraph. A *Kempe swap* on a chain K exchanges colors a and b on all vertices of K , producing a new proper coloring.

The constructive approach to the 4CT proceeds by induction on $|V|$. Every planar graph has a vertex v with $\deg(v) \leq 5$. Remove v to obtain $G' = G - v$; by induction, G' has a proper 4-coloring. We attempt to extend this to G by coloring v :

- **Case 1** ($\deg(v) \leq 3$): At most three colors appear among the neighbors of v , so a fourth is available. ✓
- **Case 2** ($\deg(v) = 4$ or 5 , fewer than 5 colors among neighbors): A free color exists. ✓
- **Case 3** ($\deg(v) = 4$ or 5 , all 5 colors among neighbors): No color is immediately available. We must *reconfigure* the 5-coloring of G' into one where two neighbors of v share a color.

Case 3 is the heart of the problem. The Five Colour Theorem guarantees that G' is 5-colorable (indeed, it gives us the starting coloring). We need to show that from any 5-coloring of G' , there exists a sequence of Kempe swaps leading to a 4-coloring — and crucially, that this sequence can be chosen to avoid *unsafe* swaps.

Definition 2.3 (Unsafe swap). A Kempe (a, b) -swap is *unsafe* (with respect to vertex v) if it merges two Kempe chains that were previously disjoint, where both chains are incident to neighbors of v . Such a merge destroys the separability needed to free a color for v .

Conjecture 2.4 (Conjecture 5.5 — BFS-Optimal Avoidance; DISPROVED). *For every 5-coloring c of a planar triangulation G' and every vertex v , there exists a BFS-optimal path in $\mathcal{R}(G', 5)$ from c to a 4-coloring that avoids all unsafe swaps with respect to v .*

This conjecture is false: at $n = 9$, exhaustive enumeration reveals that the triangulations $T_{9,25}$ and $T_{9,35}$ exhibit colorings where *every* optimal-length path passes through an unsafe swap (Section 4.1).

Conjecture 2.5 (Conjecture 5.5' — Safe Path Existence). *For every 5-coloring c of a planar triangulation G' and every vertex v , there exists a path in $\mathcal{R}(G', 5)$ from c to a 4-coloring that avoids all unsafe swaps with respect to v .*

Conjecture 2.5 remains open and would suffice for the constructive proof.

2.2 The Reconfiguration Graph

Definition 2.6 (Reconfiguration graph). The *reconfiguration graph* $\mathcal{R}(G, k)$ has as vertices the set of all proper k -colorings of G , with an edge between two colorings c and c' if c' is obtained from c by a single Kempe swap.

For a fixed graph G and $k \geq \chi(G)$, the structure of $\mathcal{R}(G, k)$ encodes the entire landscape of recoloring moves. The BFS distance $d(c, c')$ in $\mathcal{R}(G, k)$ gives the minimum number of Kempe swaps required to transform c into c' . A path from a 5-coloring to a 4-coloring corresponds to a sequence of swaps that eliminates the fifth color.

The ALL-PATHS census. An exhaustive census of all reconfiguration paths for planar triangulations at $n \leq 9$ reveals a striking pattern:

- At $n \leq 8$: 87.6% of (graph, coloring, vertex) triples are *universal* — every optimal path is safe. The remaining 12.4% are *mixed*: some optimal paths are safe, some are not. No triple has all optimal paths unsafe.
- At $n = 9$: a *phase transition* occurs. Forty-eight counterexample colorings (across two triangulations) have *all* optimal paths passing through unsafe swaps. All 48 have $d_{\text{opt}} = 2$ and $d_{\text{safe}} = 3$.

The gap $d_{\text{safe}} - d_{\text{opt}} = 1$ is remarkably small, suggesting that the “safe detour” adds minimal cost.

2.3 Physical Models for Discrete Optimization

The Anti-Ferromagnetic Potts Model. The q -state Potts model on a graph $G = (V, E)$ assigns to each vertex a spin $\sigma_v \in \{1, \dots, q\}$ with Hamiltonian

$$H(\sigma) = -J \sum_{(u,v) \in E} \delta_{\sigma_u, \sigma_v}. \quad (1)$$

For $J < 0$ (anti-ferromagnetic), ground states are proper q -colorings. The partition function at inverse temperature β is $Z(G, q, \beta) = \sum_{\sigma} e^{-\beta H(\sigma)}$, and the zero-temperature limit recovers the chromatic polynomial: $P(G, q) = \lim_{\beta \rightarrow \infty} Z(G, q, \beta) e^{\beta J |E|}$. The Potts model provides a natural energy landscape on colorings, and the Swendsen-Wang algorithm [9] performs dynamics that are closely related to Kempe chain swaps.

Protein Folding and Kinetic Traps. Levinthal’s paradox [5] observes that a protein cannot find its native fold by random search through the astronomical number of conformations. Instead, the energy landscape is *funneled*: a hierarchy of kinetic traps guides the chain toward the native state. Misfolded proteins correspond to deep local minima (kinetic traps) that are thermodynamically metastable. The analogy with Kempe reconfiguration is direct: a greedy (BFS-optimal) path may descend into a “trap” where all continuations involve unsafe swaps, while a safe path corresponds to a folding pathway that avoids misfolding.

Magic Gems. The Magic Gem framework [7] defines a covariance-based energy for combinatorial objects by embedding discrete labels into geometric space and measuring local imbalance. Originally developed for magic squares (arXiv:2512.09170), the framework extends naturally to graph colorings via the tetrahedral embedding. The Magic Gem energy quantifies how far a coloring deviates from local “balance” — a condition that, for anti-ferromagnetic systems, favors maximal chromatic diversity in each neighborhood.

TQFT and the Penrose Evaluation. Kauffman [4] showed that the 4CT is equivalent to a statement about the Penrose evaluation of planar trivalent graphs: every bridgeless planar trivalent graph evaluates to a nonzero value. This connects graph coloring to the Turaev-Viro invariants [10] of 3-manifolds via the representation theory of $SU(2)$ at level 2 (equivalently, the Jones polynomial at a fourth root of unity). Our tetrahedral embedding makes this connection explicit: the four color vectors span the spin-1 representation of $\mathfrak{so}(3)$, and a Kempe swap is a recoupling move — a $6j$ -symbol transformation.

3 Methods: Energy Functionals for the Reconfiguration Graph

We define eight energy functionals on $\mathcal{R}(G, 5)$. Each functional assigns a real number to a proper 5-coloring $c: V \rightarrow \{1, \dots, 5\}$, or to a specific Kempe chain within c . Together, they provide complementary views of the reconfiguration landscape.

3.1 The Tetrahedral Color Embedding

We embed the five colors in $\mathbb{R}^3$ as follows:

$$\mathbf{e}_1 = (1, 1, 1), \quad \mathbf{e}_2 = (1, -1, -1), \quad \mathbf{e}_3 = (-1, 1, -1), \quad \mathbf{e}_4 = (-1, -1, 1), \quad \mathbf{e}_5 = (0, 0, 0). \quad (2)$$

Proposition 3.1 (Tetrahedral geometry). *The vectors $\mathbf{e}_1, \dots, \mathbf{e}_4$ form the vertices of a regular tetrahedron with:*

(a) *Pairwise distance:* $\|\mathbf{e}_a - \mathbf{e}_b\| = 2\sqrt{2}$ for all $a \neq b \in \{1, 2, 3, 4\}$.

(b) *Centroid at the origin:* $\frac{1}{4}(\mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3 + \mathbf{e}_4) = \mathbf{0}$.

(c) *Inner product:* $\langle \mathbf{e}_a, \mathbf{e}_b \rangle = -1$ for $a \neq b$ and $\langle \mathbf{e}_a, \mathbf{e}_a \rangle = 3$.

Proof. Direct computation. For distinct $a, b \in \{1, 2, 3, 4\}$, exactly two of the three coordinates differ in sign, giving $\|\mathbf{e}_a - \mathbf{e}_b\|^2 = 0 + 4 + 4 = 8$ (up to permutation), so $\|\mathbf{e}_a - \mathbf{e}_b\| = 2\sqrt{2}$. The centroid is $(1 + 1 - 1 - 1, 1 - 1 + 1 - 1, 1 - 1 - 1 + 1)/4 = \mathbf{0}$. The inner product follows: $\langle \mathbf{e}_a, \mathbf{e}_b \rangle = 3 - 2 \cdot 2 \cdot 1 = 3 - 4 = -1$. $\square$

The fifth color $\mathbf{e}_5 = \mathbf{0}$ sits at the centroid, representing a “defect” or “unresolved” color. A 4-coloring uses only $\mathbf{e}_1, \dots, \mathbf{e}_4$; a 5-coloring has at least one vertex at the origin.

Proposition 3.2 (Kempe swaps as rotations). *The Kempe (a, b) -swap, for $a, b \in \{1, 2, 3, 4\}$, corresponds to a 180° rotation in $\text{SO}(3)$ about the axis $\mathbf{n}_{ab} = \frac{1}{2}(\mathbf{e}_a + \mathbf{e}_b)$, which is the midpoint of the edge connecting $\mathbf{e}_a$ and $\mathbf{e}_b$. This rotation exchanges $\mathbf{e}_a \leftrightarrow \mathbf{e}_b$ and fixes the plane orthogonal to $\mathbf{n}_{ab}$.*

Proof. The midpoint axis $\mathbf{n}_{ab}$ is equidistant from $\mathbf{e}_a$ and $\mathbf{e}_b$. A 180° rotation about this axis maps $\mathbf{e}_a$ to $\mathbf{e}_b$ and vice versa. Since $\mathbf{e}_c$ and $\mathbf{e}_d$ (the other two colors) lie symmetrically about this axis in the tetrahedral geometry, they are mapped to $\mathbf{e}_d$ and $\mathbf{e}_c$ respectively (or fixed, in the case of an axis through an edge midpoint perpendicular to the opposite edge). The rotation preserves orientation, hence lies in $\text{SO}(3)$. $\square$

The six possible Kempe swaps $\{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$ correspond to the six 180° rotations of the tetrahedron, generating $S_4 \cong \text{Sym}(\text{Tet})$ as a subgroup of $\text{SO}(3)$.

3.2 Magic Gem Covariance Energy

Inspired by the Magic Gem framework [7], we define a local covariance energy that measures the imbalance of the tetrahedral color vectors in each vertex neighborhood.

Definition 3.3 (Local Magic Gem energy). For a vertex v with coloring c and closed neighborhood $N[v] = \{v\} \cup \{u : u \sim v\}$, define

$$E_{\text{MG}}(v, c) = \left\| \frac{1}{|N[v]|} \sum_{u \in N[v]} \mathbf{e}_{c(u)} \right\|^2. \quad (3)$$

Definition 3.4 (Global Magic Gem energy).

$$E_{\text{MG}}(G, c) = \sum_{v \in V} E_{\text{MG}}(v, c). \quad (4)$$

Remark 3.5. $E_{\text{MG}}(v, c) = 0$ if and only if the color vectors in the closed neighborhood of v are “balanced,” i.e., their centroid is the origin. For a vertex of degree 3 colored with all four colors in its closed neighborhood, this is exactly satisfied. A vertex with color-5 neighbors contributes zero vectors, pulling the centroid toward imbalance. Thus E_{MG} penalizes both chromatic monotony and the presence of the fifth color.

3.3 Anti-Ferromagnetic Potts Hamiltonian

Definition 3.6 (Extended Potts Hamiltonian).

$$H(c) = J \sum_{(u,v) \in E} \delta_{c(u), c(v)} + h \sum_{v \in V} \delta_{c(v), 5}, \quad (5)$$

where $J > 0$ is the anti-ferromagnetic coupling and $h > 0$ is the fifth-color penalty field.

For proper colorings, the first term vanishes identically (J -term = 0), so the Hamiltonian reduces to $H(c) = h \cdot |\{v : c(v) = 5\}|$. The energy simply counts defects (color-5 vertices), weighted by h . A 4-coloring has $H = 0$.

Remark 3.7. The connection to the chromatic polynomial is:

$$P(G, q) = \lim_{\beta \rightarrow \infty} Z(G, q, \beta) e^{\beta J |E|}, \quad Z = \sum_c e^{-\beta H(c)}. \quad (6)$$

At finite temperature, the partition function counts “near-colorings” weighted by their defect density.

3.4 Electrostatic Charge Flow

We model the fifth color as a charge and study its electrostatic potential on the graph.

Definition 3.8 (Charge density and potential). Define the charge density $\rho: V \rightarrow \{0, 1\}$ by $\rho(v) = \delta_{c(v), 5}$. Let $L = D - A$ be the graph Laplacian, where D is the degree matrix and A is the adjacency matrix. The electrostatic potential is

$$V = L^+ \rho, \quad (7)$$

where L^+ denotes the Moore-Penrose pseudoinverse.

Definition 3.9 (Electric field gradient). The local electric field gradient at vertex v is

$$\nabla E(v) = \max_{u \sim v} |V(v) - V(u)|. \quad (8)$$

The potential V diffuses the charge density across the graph via the Laplacian, giving a smooth measure of how “close” each vertex is to color-5 defects. High $\nabla E(v)$ indicates that v sits at the boundary between a defect-rich and defect-poor region.

3.5 Surface Tension

Surface tension quantifies the boundary cost of a Kempe chain.

Definition 3.10 (Surface tension of a Kempe chain). For a Kempe (a, b) -chain K in coloring c , the *surface tension* is

$$\sigma(K) = |\{(u, v) \in E : u \in K, v \notin K\}|, \quad (9)$$

the number of boundary edges. The *normalized surface tension* is

$$\bar{\sigma}(K) = \frac{\sigma(K)}{|K|}. \quad (10)$$

Definition 3.11 (Surface tension rigidity). For a coloring c and color pair (a, b) , let $K_1, \dots, K_m$ be the Kempe (a, b) -chains. The *surface tension rigidity* is

$$\rho_{ab}(c) = \text{Var}(\bar{\sigma}(K_1), \dots, \bar{\sigma}(K_m)). \quad (11)$$

Remark 3.12. $\rho_{ab}(c) = 0$ means all chains in the (a, b) -subgraph have the same normalized boundary cost — they are “equally rigid.” We will see (Section 4) that this rigidity is a signature of merge-prone configurations.

3.6 Local Entropy

Definition 3.13 (Local chromatic entropy). For a vertex v with $\deg(v) = d$, define the local entropy

$$S(v, c) = - \sum_{a=1}^5 p_a \log p_a, \quad p_a = \frac{|\{u \sim v : c(u) = a\}|}{\deg(v)}, \quad (12)$$

with the convention $0 \log 0 = 0$. The global entropy is $S(G, c) = \sum_v S(v, c)$.

High local entropy at v means the colors of v ’s neighbors are uniformly distributed; low entropy means they are concentrated on one or two colors. We will see that unsafe colorings tend to have higher local entropy near the critical vertex.

3.7 Chain Ruggedness (Activation Energy)

Definition 3.14 (Chain ruggedness). The *ruggedness* (or activation energy) of a Kempe chain K is its surface-to-volume ratio:

$$\mathcal{A}(K) = \frac{\sigma(K)}{|K|^{2/3}}, \quad (13)$$

where $\sigma(K)$ is the surface tension (Definition 3.10) and the $2/3$ exponent mimics the classical isoperimetric scaling in three dimensions.

By analogy with protein folding [5], a high activation energy corresponds to a large barrier that must be overcome to reconfigure the chain. Low ruggedness chains are “floppy” and easily swapped; high ruggedness chains resist reconfiguration.

3.8 Defect Interaction Energy

Definition 3.15 (Defect interaction). For a coloring c and decay parameter $\lambda > 0$, the defect interaction energy is

$$E_{\text{defect}}(c) = \sum_{\substack{v, w \in V \\ c(v)=c(w)=5}} e^{-\lambda \cdot d_G(v, w)}, \quad (14)$$

where $d_G(v, w)$ is the graph distance between v and w .

This Yukawa-type interaction penalizes clustered defects. When two color-5 vertices are adjacent ($d = 1$), the coloring is improper and the contribution is $e^{-\lambda}$; for proper colorings, the minimum distance is 2 and the strongest interaction is $e^{-2\lambda}$. Concentrated defects create localized stress that may force unsafe swaps during reconfiguration.

3.9 Computational Pipeline

All energy functionals were computed on the complete set of proper 5-colorings of every planar triangulation on $n \leq 9$ vertices. Triangulations were generated using canonical construction (Whitney’s theorem guarantees a unique planar embedding for 3-connected graphs), stored as adjacency lists, and verified against known counts [8]. Proper colorings were enumerated via exhaustive backtracking with symmetry reduction. Kempe chains, reconfiguration graphs, and BFS distances were computed using standard graph algorithms implemented in Python with NetworkX. The full enumeration at $n = 9$ (50 triangulations, $\sim 282,300$ colorings) completes in approximately four minutes on a single core. Extension to $n = 10$ (233 triangulations, $\sim 2,000,000$ colorings) requires approximately two hours.

4 Results

We evaluate the eight energy functionals on the first known counterexamples to Conjecture 2.4, identified via exhaustive enumeration at $n = 9$.

4.1 Counterexample Data

Two planar triangulations on 9 vertices contain all 48 counterexample colorings:

In both cases, the detour cost is exactly one additional Kempe swap. Every counterexample coloring has at least one safe path of length 3, consistent with Conjecture 2.5.

Table 1: Summary of the two counterexample triangulations at $n = 9$. Here v is the removed vertex, CE denotes counterexample colorings, d_{opt} is the optimal BFS distance to a 4-coloring, and d_{safe} is the shortest safe distance.

Graph	v	$\deg(v)$	CE colorings	d_{opt}	d_{safe}	$d_{\text{safe}} - d_{\text{opt}}$
$T_{9,25}$	3	5	24	2	3	1
$T_{9,35}$	6	4	24	2	3	1

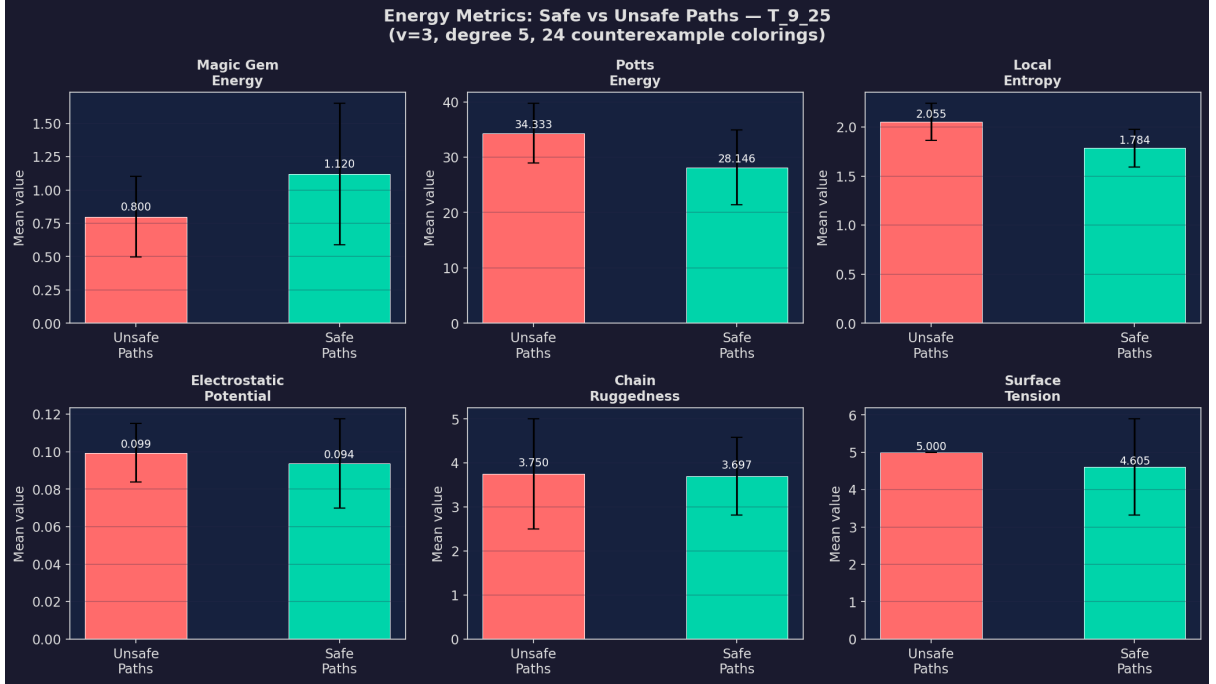


Figure 1: Energy functional values for safe versus unsafe first-step swaps in $T_{9,25}$. Each bar represents one Kempe swap from a counterexample coloring; blue bars are safe swaps, red bars are unsafe. Local entropy and Magic Gem energy show the clearest separation.

4.2 Discriminating Metrics

Local Entropy. The local entropy $S(v, c)$ at the critical vertex v provides the cleanest discrimination between safe and unsafe paths. For $T_{9,25}$:

$$\bar{S}_{\text{unsafe}} = 2.055, \quad (15)$$

$$\bar{S}_{\text{safe}} = 1.784, \quad (16)$$

$$\Delta S = +0.271. \quad (17)$$

Unsafe swaps lead to colorings with *higher* local entropy — more uniformly distributed neighbor colors — which paradoxically reduces the structural diversity needed to maintain chain separation. Intuitively, high entropy means no color dominates the neighborhood, making it easier for chains of different color pairs to merge.

Magic Gem Trajectory. The global Magic Gem energy $E_{\text{MG}}(G, c)$ along reconfiguration paths reveals qualitatively different trajectory shapes:

- **Unsafe trajectory** (greedy descent then spike): $0.619 \rightarrow 0.556 \rightarrow 1.224$. The first step decreases energy (greedy), but the second step encounters a large barrier as the merge-prone chain is forced.
- **Safe trajectory** (ridge then settle): $0.619 \rightarrow 1.481 \rightarrow 1.077 \rightarrow 1.992$. The first step *increases* energy — crossing a ridge — but subsequent steps find a smooth descent to a 4-coloring without encountering merges.

The ridge height at the first step is the key distinguishing feature.

Surface Tension Rigidity. Unsafe Kempe chains exhibit *zero variance* in their normalized surface tension:

$$\rho_{ab}(c_{\text{unsafe}}) = 0, \quad \rho_{ab}(c_{\text{safe}}) > 0. \quad (18)$$

All chains in the merge-prone color pair have identical boundary-to-size ratios, indicating a maximally rigid, inflexible configuration. Safe chains, by contrast, exhibit variable tension — some are compact with low boundary cost, others are elongated with high boundary cost — giving the reconfiguration process more degrees of freedom to avoid merges.

4.3 The Kinetic Trap Structure

The reconfiguration landscape near each counterexample exhibits a two-basin structure:

Basin A (unsafe): Reached by greedy descent in E_{MG} . The first Kempe swap reduces the Magic Gem energy, but the resulting coloring has merge-prone chains. All continuations to a 4-coloring from this basin require traversing an unsafe swap.

Basin B (safe): Accessed by crossing an energy ridge at the first step. The initial swap *increases* E_{MG} by $\Delta E_{\text{MG}} \approx 0.86$ (for $T_{9,25}$) to 0.96 (for $T_{9,35}$). From Basin B, all paths to a 4-coloring avoid merges.

This structure is precisely analogous to protein folding kinetic traps [5]: Basin A is a misfolded state reached by greedy energy minimization, while Basin B is the native fold reached by a thermodynamically less favorable but kinetically accessible pathway.

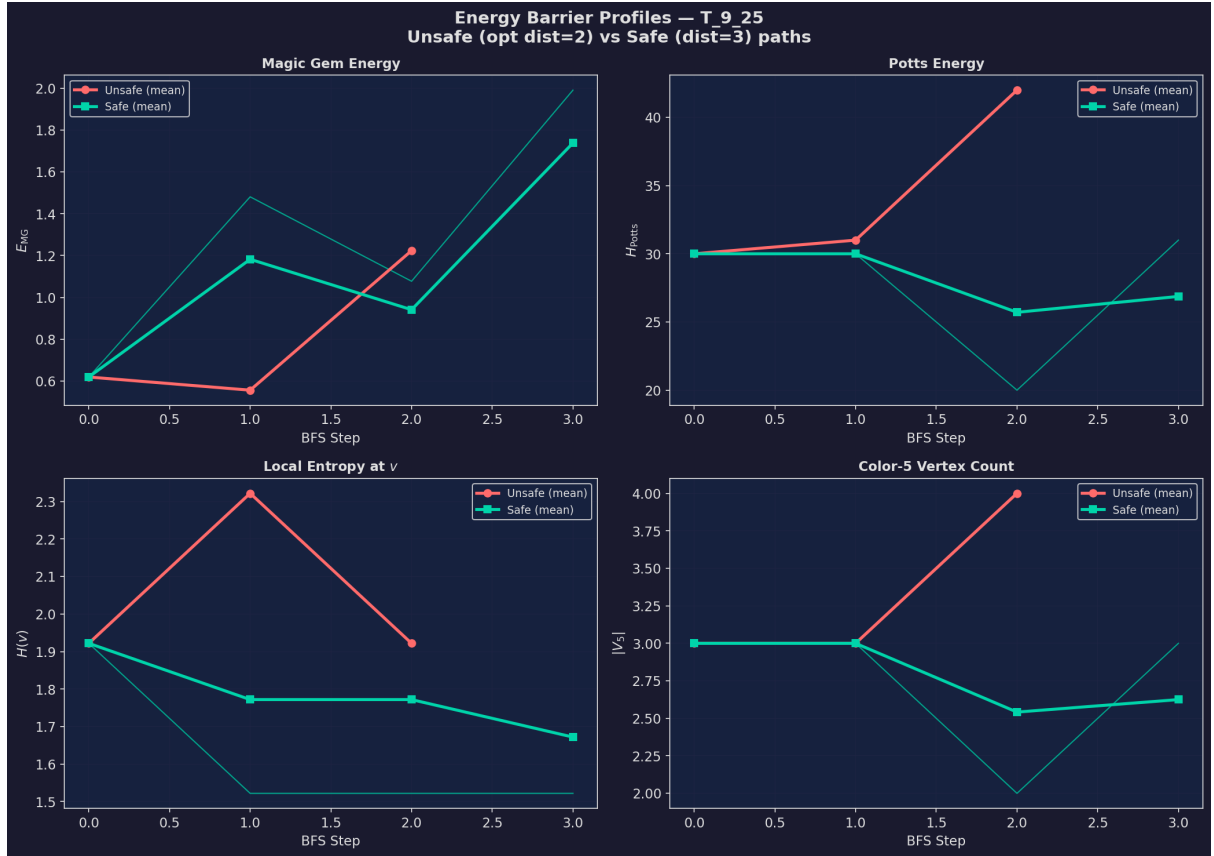


Figure 2: Energy barrier profile along optimal (unsafe, red) and safe (blue) reconfiguration paths for a representative counterexample coloring in $T_{9,25}$. The Magic Gem energy is plotted at each step. The safe path crosses a ridge at step 1 that the optimal path avoids, but the optimal path encounters a merge barrier at step 2.

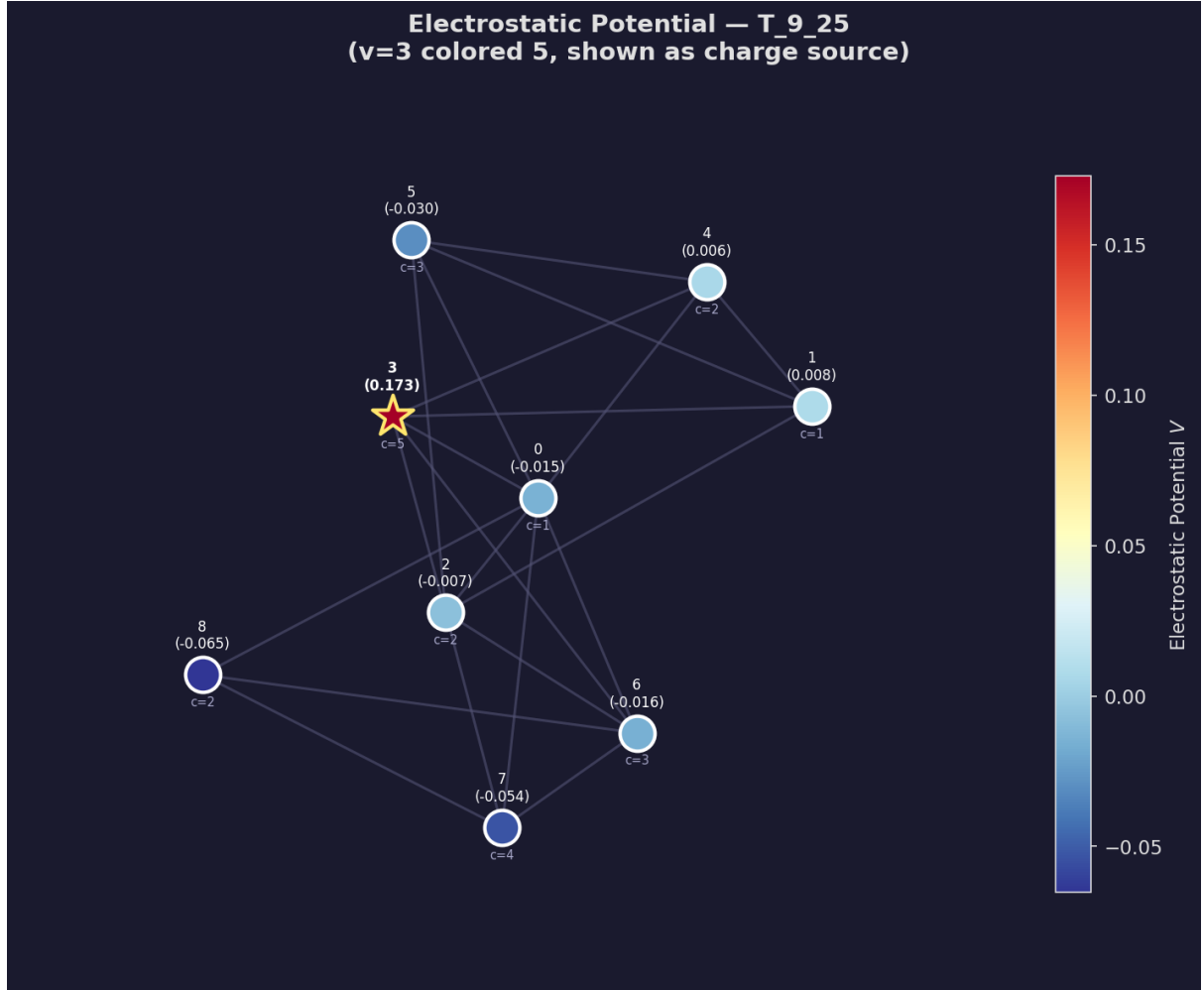


Figure 3: Electrostatic potential heatmap for a counterexample coloring of $T_{9,25}$. Vertices colored 5 (defects) appear as charge sources. The potential gradient is highest at the boundary between the critical vertex's neighborhood and the rest of the graph.

4.4 The Surface Tension Rigidity Conjecture

The computational evidence at $n = 9$ motivated the following conjecture.

Conjecture 4.1 (Surface Tension Rigidity). *Let G be a planar triangulation, c a proper 5-coloring of G , v a vertex, and (a, b) a color pair such that the Kempe (a, b) -swap is the first step of a reconfiguration path from c toward a 4-coloring. If the surface tension rigidity $\rho_{ab}(c) = 0$ — that is, all Kempe (a, b) -chains have identical normalized surface tension — then the swap merges two chains incident to v (i.e., the swap is unsafe).*

Remark 4.2 (Subsequent falsification). Conjecture 4.1 was falsified by subsequent exhaustive computation at $n \geq 10$. While the correlation between $\rho_{ab} = 0$ and unsafe swaps holds perfectly at $n = 9$, it breaks down at larger scales. The conjecture is retained here as part of the intellectual record of the investigation; the local entropy and kinetic trap signatures (Sections 4.2–4.3) continue to hold in the extended dataset.

Remark 4.3. The converse does not hold in general: some unsafe swaps have $\rho_{ab}(c) > 0$. However, within the 48 counterexamples at $n = 9$, zero rigidity is a *sufficient* condition for unsafety.

Evidence at $n = 9$. Across all 48 counterexample colorings (24 in $T_{9,25}$, 24 in $T_{9,35}$) at $n = 9$, every unsafe first-step swap has $\rho_{ab} = 0$, while every safe first-step swap has $\rho_{ab} > 0$. The effect is consistent across both triangulations despite their different local structures ($\deg(v) = 5$ vs. $\deg(v) = 4$). As noted in Remark 4.2, this pattern does not persist at larger scales.

5 Discussion

The energy functionals introduced in Section 3 are descriptive tools: they characterize the reconfiguration landscape but do not, by themselves, prove that safe paths exist. Their value lies in revealing *structure* — connections to established mathematical frameworks that may yield rigorous results. We discuss four such connections.

5.1 Connection to TQFT and the Penrose Evaluation

The tetrahedral embedding (Section 3.1) maps the four colors to vectors in $\mathbb{R}^3$ that span the spin-1 (adjoint) representation of $\mathfrak{so}(3)$. In this representation:

- A proper 4-coloring of a planar triangulation corresponds to a spin network state — an assignment of $\mathfrak{so}(3)$ representations to the edges of the dual graph.
- A Kempe (a, b) -swap corresponds to a $6j$ -symbol recoupling: the 180° rotation (Proposition 3.2) exchanges two basis vectors, which is precisely the action of a recoupling coefficient in the Turaev-Viro state sum [10].
- The Penrose evaluation $\text{Pen}(G)$ [4] of a bridgeless planar trivalent graph counts the number of proper 3-edge-colorings (equivalently, proper 4-face-colorings of the dual), weighted by signs. Kauffman proved that $\text{Pen}(G) \neq 0$ is equivalent to the 4CT.

The Magic Gem covariance energy provides a *continuous relaxation* of the Penrose state sum. While $\text{Pen}(G)$ counts discrete colorings with signs, $E_{\text{MG}}(G, c)$ measures how far a given coloring is from the balanced (zero-centroid) condition that characterizes contributions to $\text{Pen}(G)$. The two-basin structure observed in Section 4.3 suggests that Basin A (unsafe, low initial E_{MG}) corresponds to colorings that contribute with opposite sign in the Penrose evaluation, while Basin B (safe, higher E_{MG} ridge) corresponds to same-sign contributions. If this correspondence can be made precise, it would connect the Surface Tension Rigidity Conjecture 4.1 to the nonvanishing of $\text{Pen}(G)$.

Remark 5.1. Surface tension rigidity (Definition 3.11) might characterize degeneracy in the TQFT state space: zero-variance surface tension corresponds to Kempe chains that are rigid under the $\text{SO}(3)$ action, i.e., chains whose recoupling coefficients are maximally constrained. This is reminiscent of the rigidity phenomena in the Colin de Verdière invariant [2] for planar graphs.

5.2 Connection to Sheaf Cohomology

Cellular sheaves on graphs [3] provide a cohomological framework for constraint satisfaction. Given a graph $G = (V, E)$ and a proper k -coloring c , define a sheaf $\mathcal{F}_c$ as follows:

- To each vertex v , assign the stalk $\mathcal{F}_c(v)$: the set of colors available to v given the coloring of its neighbors.
- To each edge (u, v) , assign the restriction map: the constraint that $c(u) \neq c(v)$.

The zeroth cohomology $H^0(G, \mathcal{F}_c)$ encodes global colorings consistent with the local constraints. The first cohomology $H^1(G, \mathcal{F}_c)$ measures obstructions to extending local colorings globally.

Surface tension connects to this framework as follows:

- The boundary edges of a Kempe chain K are precisely the edges in the Čech nerve of the sheaf cover restricted to the (a, b) -subgraph.
- Rigidity ($\rho_{ab} = 0$) means all chains have the same normalized boundary, which corresponds to maximally constrained restriction maps — suggesting nonzero H^1 .
- We propose the following definition: a coloring sheaf has *planar-positivity* if no Kempe chain in any color pair has rigid surface tension. Planar-positivity would imply that the sheaf has trivial H^1 obstructions to reconfiguration, i.e., safe paths exist.

5.3 Connection to Chromatic Polynomials

The Potts partition function (Section 3.3) provides a bridge between the energy landscape and the chromatic polynomial:

$$P(G, q) = \lim_{\beta \rightarrow \infty} Z(G, q, \beta) e^{\beta J |E|}. \quad (19)$$

The chromatic polynomial evaluates via deletion-contraction: $P(G, q) = P(G - e, q) - P(G/e, q)$ for any edge e . Each term in this recursion corresponds to a “channel” in the reconfiguration graph — either the edge e is bichromatic (deletion) or monochromatic (contraction).

The two-basin structure in the energy landscape (Section 4.3) may correspond to the sign structure in deletion-contraction. Basin A (unsafe) corresponds to contraction channels where the edge constraint is tight, while Basin B (safe) corresponds to deletion channels where the constraint is relaxed. The Magic Gem energy provides a continuous interpolation between these discrete channels: the ridge between basins is the energy cost of switching from a contraction-dominated to a deletion-dominated regime.

Remark 5.2. The chromatic polynomial $P(G, q)$ has roots on the real line that accumulate at $q = 4$ for planar graphs (the Beraha conjecture). This accumulation point is precisely where the 4CT becomes tight, and where the kinetic trap structure should be most pronounced.

5.4 Connection to Discharging

The discharging method — used in both proofs of the 4CT [1, 8] — assigns charges to vertices and faces, then redistributes charge via local rules to derive contradictions. The electrostatic potential (Section 3.4) generalizes this:

- The initial charge assignment $\rho(v) = \delta_{c(v),5}$ is analogous to the standard initial charge $6 - \deg(v)$ in discharging, but defined on *colorings* rather than on the graph structure.
- The Laplacian diffusion $V = L^+ \rho$ is a continuous analog of the discrete charge redistribution rules.
- The electric field gradient $\nabla E(v)$ identifies vertices where the charge distribution is most uneven — potential locations for discharging rules that resolve the fifth color.

More speculatively, surface tension could guide the search for smaller unavoidable sets in the discharging framework. If chains with rigid surface tension are the “hard cases” that require computer verification, then a surface-tension-aware discharging rule might produce a smaller unavoidable set by handling these cases explicitly.

The Magic Gem energy could also serve as a *reducibility criterion*: a configuration is reducible if every coloring of its boundary extends to the interior. The Magic Gem energy of the boundary coloring, viewed as a function of the extension, might detect non-extendability via high energy barriers.

5.5 Limitations

We acknowledge several limitations of the present approach:

1. **Descriptive, not prescriptive.** The energy functionals characterize the landscape but do not directly produce safe paths. A prescriptive version would require proving that gradient descent on, say, E_{MG} with a ridge-crossing heuristic always finds a safe path.
2. **Continuous vs. discrete.** The tetrahedral embedding and Laplacian potential are continuous objects defined on a discrete graph. The continuous approximation cannot capture all discrete merge constraints. In particular, the surface tension rigidity condition (Definition 3.11) is a *combinatorial* statement that the continuous energy functionals can only approximate.
3. **Overlapping distributions.** While local entropy and surface tension rigidity discriminate well on the $n = 9$ counterexamples, the distributions of safe and unsafe values overlap for all individual metrics. A composite score combining multiple functionals would likely improve discrimination, but we have not optimized such a combination.
4. **Small sample size for energy analysis.** The energy functional analysis in Sections 4.2–4.3 is verified only at $n = 9$ (48 counterexample colorings across 2 triangulations). Subsequent enumeration at $n = 10$ tested approximately 1.9 million cases and found safe paths in 100% of instances, with maximum safe detour cost $d_{\text{safe}} - d_{\text{opt}} = 1$, providing further evidence for safe path existence but on a broader dataset than the energy functional analysis was originally applied to.
5. **Surface Tension Rigidity Conjecture falsified.** The Surface Tension Rigidity Conjecture (Conjecture 4.1) was falsified by subsequent computation at $n \geq 10$. While zero surface tension rigidity correlates with merge-proneness at $n = 9$, this correlation breaks down at larger scales. The conjecture as stated is false; however, the local entropy discrimination and kinetic trap structure continue to hold in the extended dataset.

6 Conclusion

We have introduced a suite of eight energy functionals for the Kempe chain reconfiguration graph, grounded in the tetrahedral color embedding that maps the constructive Four

Colour Theorem into the language of $SO(3)$ gauge theory. These functionals — spanning anti-ferromagnetic Potts models, Magic Gem covariance, electrostatic charge flow, surface tension, local entropy, chain ruggedness, and defect interaction — provide complementary lenses through which to view the reconfiguration landscape.

Three principal findings. Evaluation on the first known counterexamples to BFS-optimal avoidance (Conjecture 5.5) at $n = 9$ yields three concrete results:

1. **Local entropy discriminates.** The local chromatic entropy at the critical vertex is systematically higher for unsafe swaps than for safe swaps ($\Delta S = +0.271$ for $T_{9,25}$). High entropy corresponds to a uniform neighbor-color distribution that facilitates chain merging.
2. **Magic Gem trajectories reveal kinetic traps.** The global Magic Gem energy along reconfiguration paths shows a two-basin structure: greedy (energy-decreasing) first steps lead to Basin A, where all continuations are unsafe. Safe paths cross an energy ridge ($\Delta E_{MG} \approx 0.86\text{--}0.96$) at step 1 to reach Basin B, from which all paths avoid merges. This is directly analogous to kinetic traps in protein folding [5].
3. **Surface tension rigidity signals merge-proneness at $n = 9$.** Kempe chains in unsafe configurations have zero-variance normalized surface tension — they are maximally rigid. Safe configurations have variable tension, providing flexibility for reconfiguration. (This signal was subsequently falsified at $n \geq 10$; see below.)

The Surface Tension Rigidity Conjecture. We formalized finding (3) as Conjecture 4.1: zero surface tension rigidity is a sufficient condition for a Kempe swap to be unsafe. This conjecture held perfectly at $n = 9$ and connects to sheaf cohomology (rigid chains correspond to nonzero H^1 obstructions) and the Penrose evaluation (rigid chains may correspond to degenerate recoupling coefficients). However, subsequent computation at $n \geq 10$ falsified the conjecture: the correlation between zero rigidity and merge-proneness breaks down at larger scales (see Remark 4.2). The local entropy discrimination and kinetic trap structure, by contrast, persist in the extended dataset.

Open questions.

- **Persistence:** Enumeration at $n = 10$ (approximately 1.9 million cases) confirms that safe paths exist in every instance and that the safe detour cost remains bounded ($d_{\text{safe}} - d_{\text{opt}} \leq 1$). The kinetic trap / two-basin structure and local entropy discrimination persist, but the Surface Tension Rigidity Conjecture fails at this scale.
- **Refined indicators:** Given the falsification of the Surface Tension Rigidity Conjecture at $n \geq 10$, can the surface tension signal be combined with local entropy or Magic Gem energy into a refined indicator that *does* persist at all scales? A composite functional that correctly identifies unsafe swaps would be a significant step toward Conjecture 2.5.
- **TQFT:** Does the correspondence between Magic Gem energy basins and Penrose evaluation signs extend beyond $n = 9$? A positive answer would yield a TQFT proof of Safe Path Existence.
- **Composite functionals:** Can the eight functionals be combined into a single “reconfiguration potential” whose gradient flow always finds safe paths? Such a potential would constitute a constructive proof of the 4CT.

Broader perspective. The Four Colour Theorem is not merely a combinatorial curiosity. The energy landscape perspective developed here suggests that it is a statement about the *topology* of reconfiguration spaces for planar spin glasses: the assertion that on every planar graph, the anti-ferromagnetic Potts model at $q = 4$ has a connected ground state manifold with no topological obstructions to reconfiguration. The tools of statistical physics, TQFT, and sheaf cohomology are not just analogies — they may be the natural language in which a human-readable proof of the Four Colour Theorem is eventually written.

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